

importance when the data is sensitive or there are regulatory requirements.

Open source also forces innovation at a pace faster than almost any other system. The community ensures features evolve quickly, bugs are rapidly fixed, and best methods are widely propagated. This is absolutely necessary in the data management business where the requirements are real-time analytics, pipeline automation, and scalable architectures.

In addition, open source is cost-effective. Open source-licensed tools protect startups from exorbitant licence fees and do not chain them to vendors.

Understanding data management

Data management is the process of collecting, storing (schema), organising (structure), protecting (infrastructure), and using data effectively and efficiently to ensure the right data is made available to the right people, at the right time, and in the right format.

Data management is akin to a building's foundation. Without a solid measurement framework, everything

built on top of it, including analytics, AI, dashboards, and decisions, will be at risk of collapsing. Good data management leads to a valuable, credible, and usable asset leveraged to grow the business.

Role of data management in business operations

Effective data management enables companies to operate faster, smarter, and with greater precision.

Informed decision-making: Properly organised data supplies precise real-time information, which enables leaders to make the right business choices based on market trends and demand predictions. Decision-making relies on guesswork or outdated information when data is unclear and inaccessible.

Enhanced customer experience: Businesses that maintain properly organised customer data can deliver personalised services while resolving issues faster and creating deeper customer connections. A retail brand uses customer purchase history together with browsing trends to generate suitable product recommendations.

Data security

All businesses — from emerging startups to multinational corporations — must protect their data because it determines their success, public image and chances of survival.

Data security consists of the multiple components and methods that organisations use to prevent unauthorised access, damage, theft and loss of their digital information. The system protects important information including customer data, financial records, and operational data by maintaining its confidentiality and accuracy while restricting access to only approved users.

Organisations must now fulfill legal requirements that protect personal and sensitive data because data has become a highly regulated asset. Businesses that fail to meet compliance standards face substantial financial penalties and legal challenges, and suffer from diminished public confidence.

Key data protection laws include:

India: Under India's Digital Personal Data Protection (DPDP) Act, 2023, businesses must safeguard

Types of security threats: Internal, external, and emerging

Threat type	Description	Examples
Internal threats	These originate from within the organisation — either due to negligence or malicious intent.	Accidental data leaks by employees; misconfigured databases; insider attacks or privilege misuse
External threats	These come from outside in the form of attackers looking to steal, corrupt, or extort data.	Phishing attacks; ransomware; hacking and credential theft
Emerging threats	New, evolving dangers driven by technological changes and advanced attack methods.	AI-driven deepfake attacks; supply chain attacks; exploits in IoT and smart devices

Where data management and data security intersect

Aspect	Data management	Data security	Intersection
Data access	Ensures users can retrieve the data they need.	Controls who is allowed to access what data.	Role-based access controls (RBAC)
Data quality	Focuses on data accuracy and integrity.	Protects data from unauthorised alteration or corruption.	Validations, auditing, and version control
Storage and backup	Manages storage, archiving, and data recovery.	Secures backups, prevents data loss, and enables disaster recovery.	Encrypted backups with retention policies
Compliance and governance	Implements policies for ethical and legal data use.	Ensures adherence to security standards and legal obligations.	Unified data governance framework